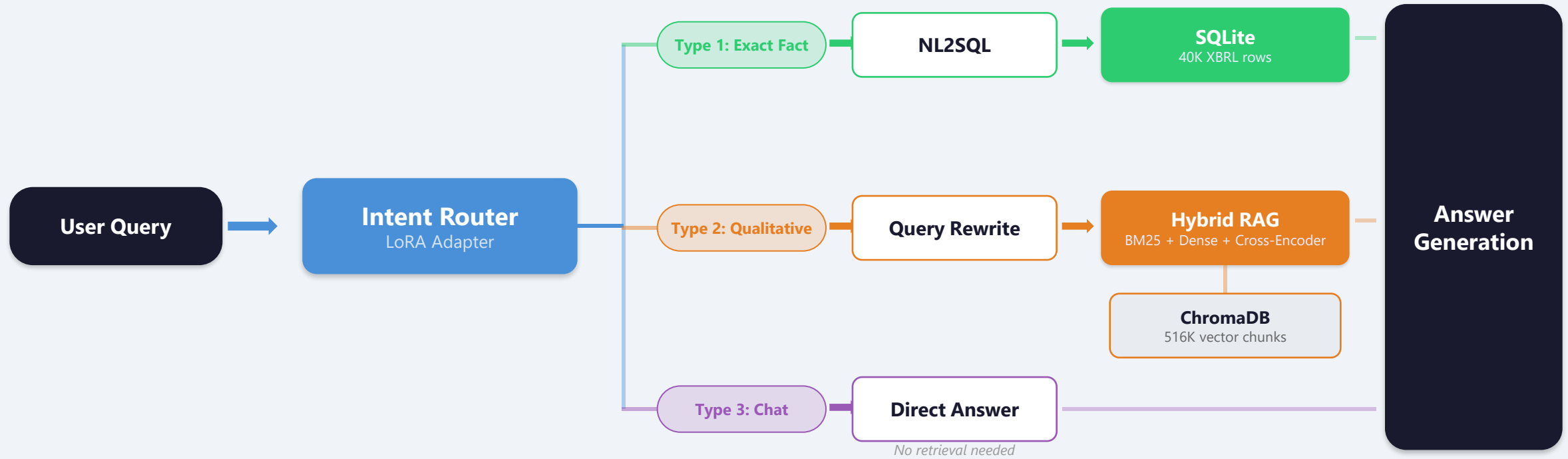


Local Deployment and Infrastructure Acceleration for a Financial QA System

Yuan Zhao | May 2026

System Architecture



92 S&P 100 tickers | 2019-2024 | ~120K text chunks + ~40K structured XBRL rows

Motivation

Accuracy:

Cloud LLM APIs (GPT-4o) have **knowledge cutoff issues** -- miss recent FY2023 data

Cloud APIs **hallucinate financial numbers** -- 42.9% value accuracy on our benchmark

Privacy:

Sending proprietary financial queries to external APIs

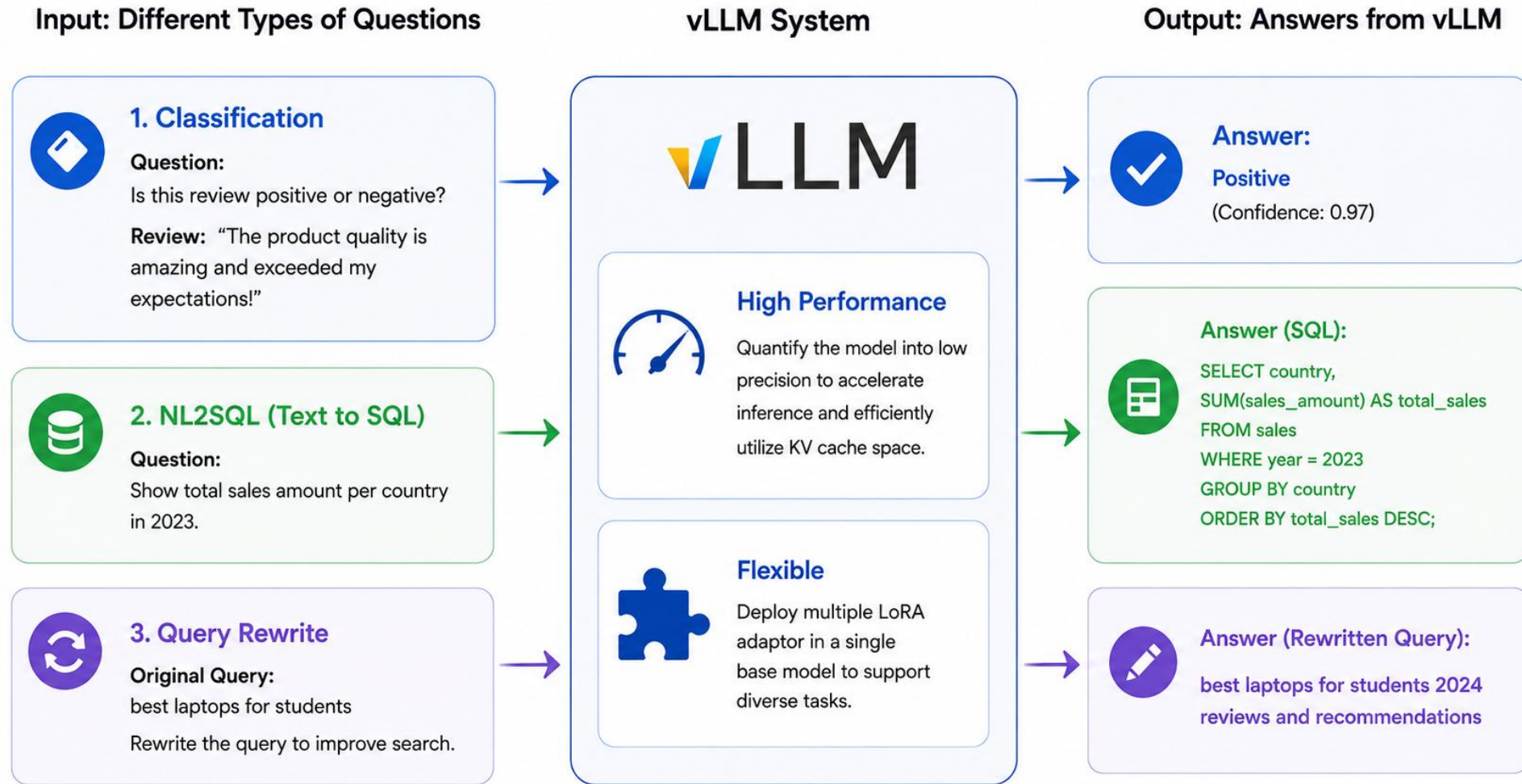
Cost:

GPT-4o: \$2.50 / 1M input tokens and \$10 / 1M output tokens

Speed:

Run Three different module in a local device with 26GB memory space is challenge.

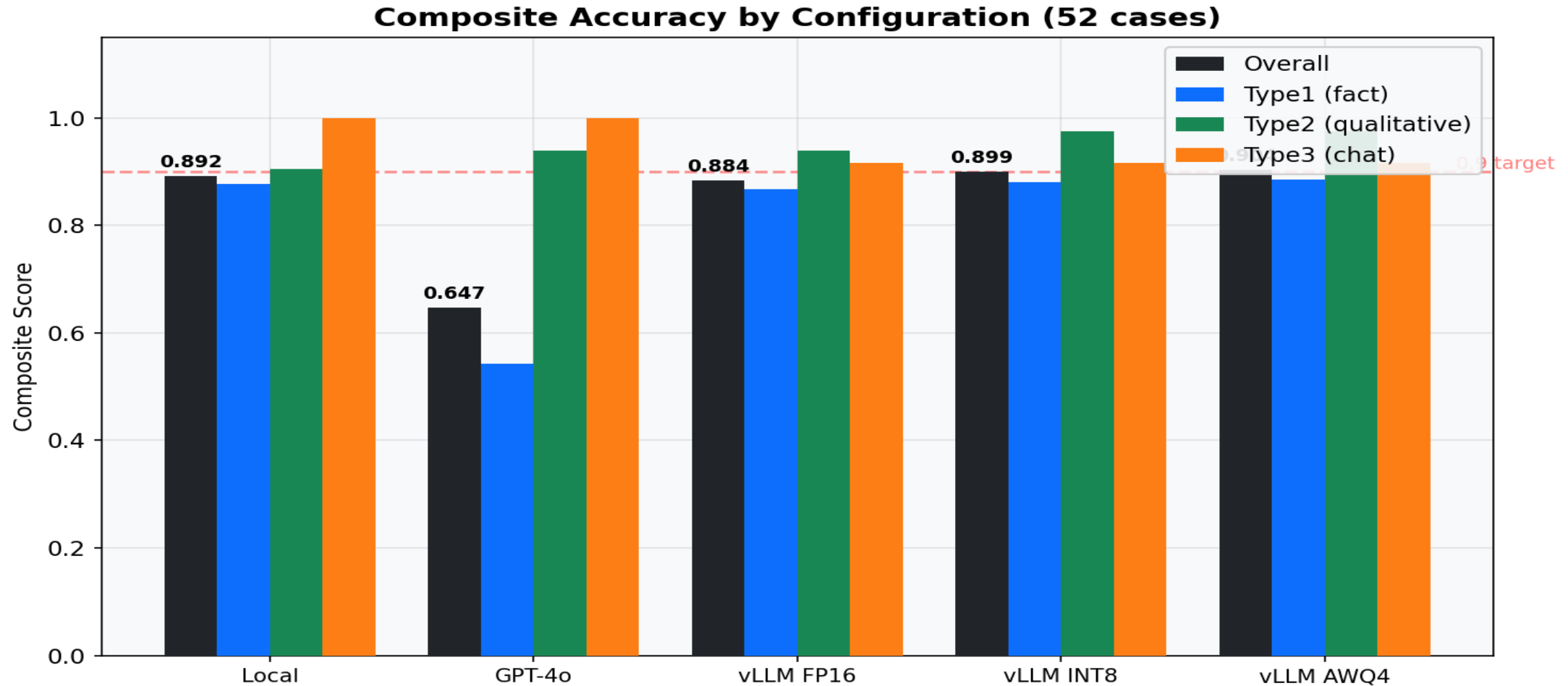
vLLM Solve the Problem:



Quantization Strategy: FP16 vs INT8 vs AWQ4

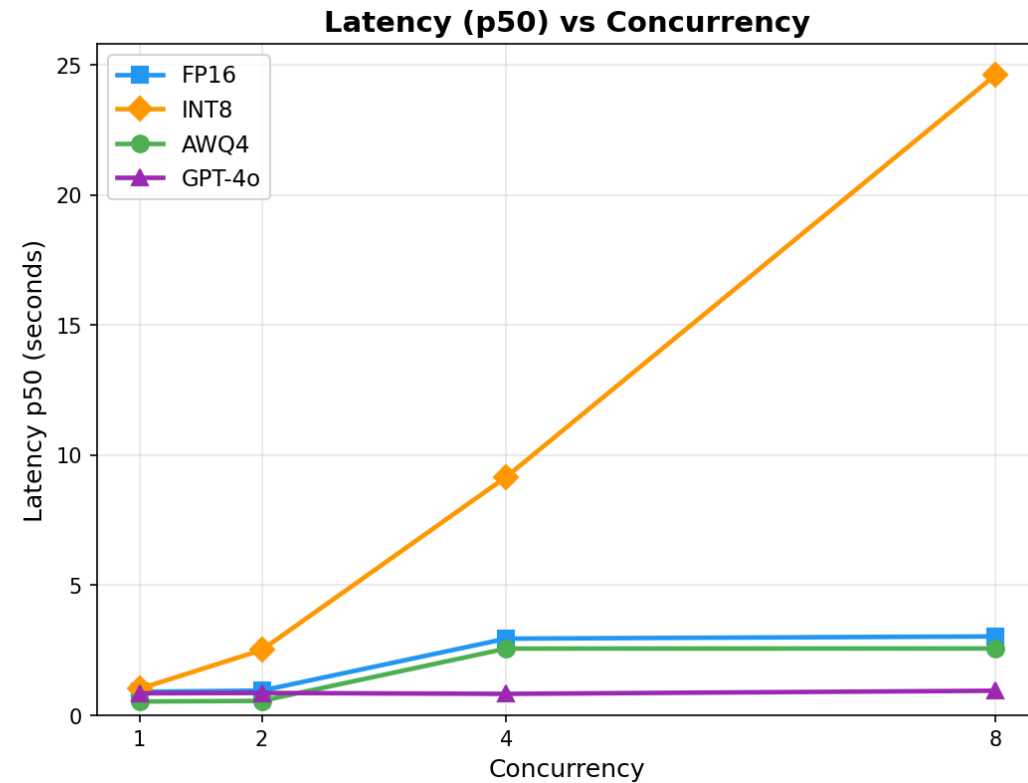
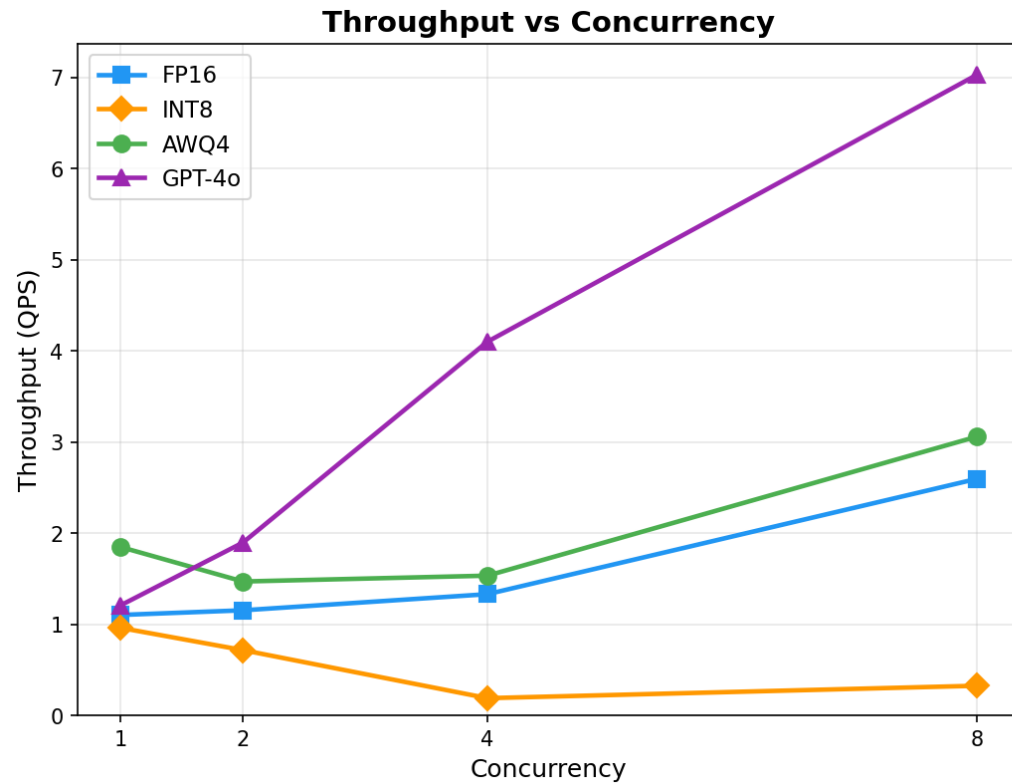
Method	Weight Size	VRAM (weights)	KV Cache Budget
FP16	16-bit	~6 GB	~18 GB
INT8	8-bit	~3 GB	~21 GB
AWQ4	4-bit	~1.5 GB	~22.5 GB

Accuracy: Unified 52-Case Benchmark Compare with GPT-4o



AWQ4 achieves the highest overall composite score (0.904), crossing the 0.9 target. GPT-4o fails on Type1 due to knowledge cutoff.

Performance: Throughput and Latency



AWQ4 have best performance in local device.

Final Results: AWQ4 vs Cloud (GPT-4o)

Metric	AWQ4 (Local)	GPT-4o (API)
Composite Accuracy	0.9038	0.6467
Value Accuracy	100%	42.9%
Intent Accuracy	100%	100%
Keyword Hit Rate	93.9%	75.7%
Fluency (1-5)	3.4	4.7
Latency p50 (c=1)	0.53s	0.85s
Peak QPS	3.46 (c=16)	7.03 (c=8)
Cost per 1K queries	~\$0	~\$15-30
Data Privacy	Fully local	Sent to OpenAI

Conclusion & Future Plan

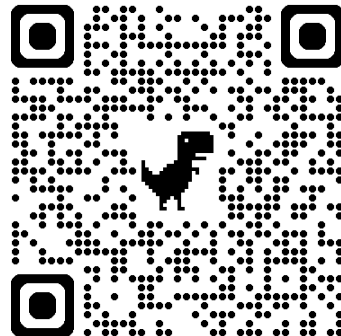
Conclusion:

A local 3B model with LoRA SFT and AWQ4 Quantization + vLLM achieves 0.90 accuracy, 100% value accuracy, and 0.53s latency -- beating GPT-4o on financial QA at \$0/query.

Future Plan:

Fluency gap – larger scale answer generator

Multi-GPU scaling -- tensor parallelism or load-balanced vLLM for >5 QPS



Github Link